

The Problem

6.8 million Americans use mobility assistive device. Yet around 10.1% of U.S adults, a staggering 19 million people, report mobility difficulty like standing limitations.

The Solution

A wearable assistive device for patients unable to stand up independently. By focusing on portability, automation and adaptive engagement of the device, we ensured compatibility with a broad range of users.

Interviews

- 1

Jixiang Feng, Stroke survivor and his primary caregiver.

Highlighted how exhausting repeated sit-to-stand transfers are at home. Emphasized the need for a simple assistive device that preserves the user’s independence.
- 2

Connie Leibow, Pediatric Physical Therapist.

Guided us to pivot away from a hip-torque exoskeleton toward a simpler, more targeted sit-to-stand solution.

Brainstorming

Actuation					
Control					
Push up interface					
Wearability & disengaging					
Seat design and attachment					

(bad) 1-5 (good) score	weight	Knee exoskeleton	Push up seat	Powered stand alone
Feasibility (10 weeks)	0.2	2	3	2
Ease of adaptability to user	0.2	4	5	4
Prototype cost (budget \$700)	0.1	3	3	1
Novelty design	0.1	3	4	2
Complexity design	0.2	4	4	2
Cost final device	0.2	3	4	2
Total score Pugh chart	1	3.2	3.9	2.3



SPROUT UP

Support You Can Carry With You

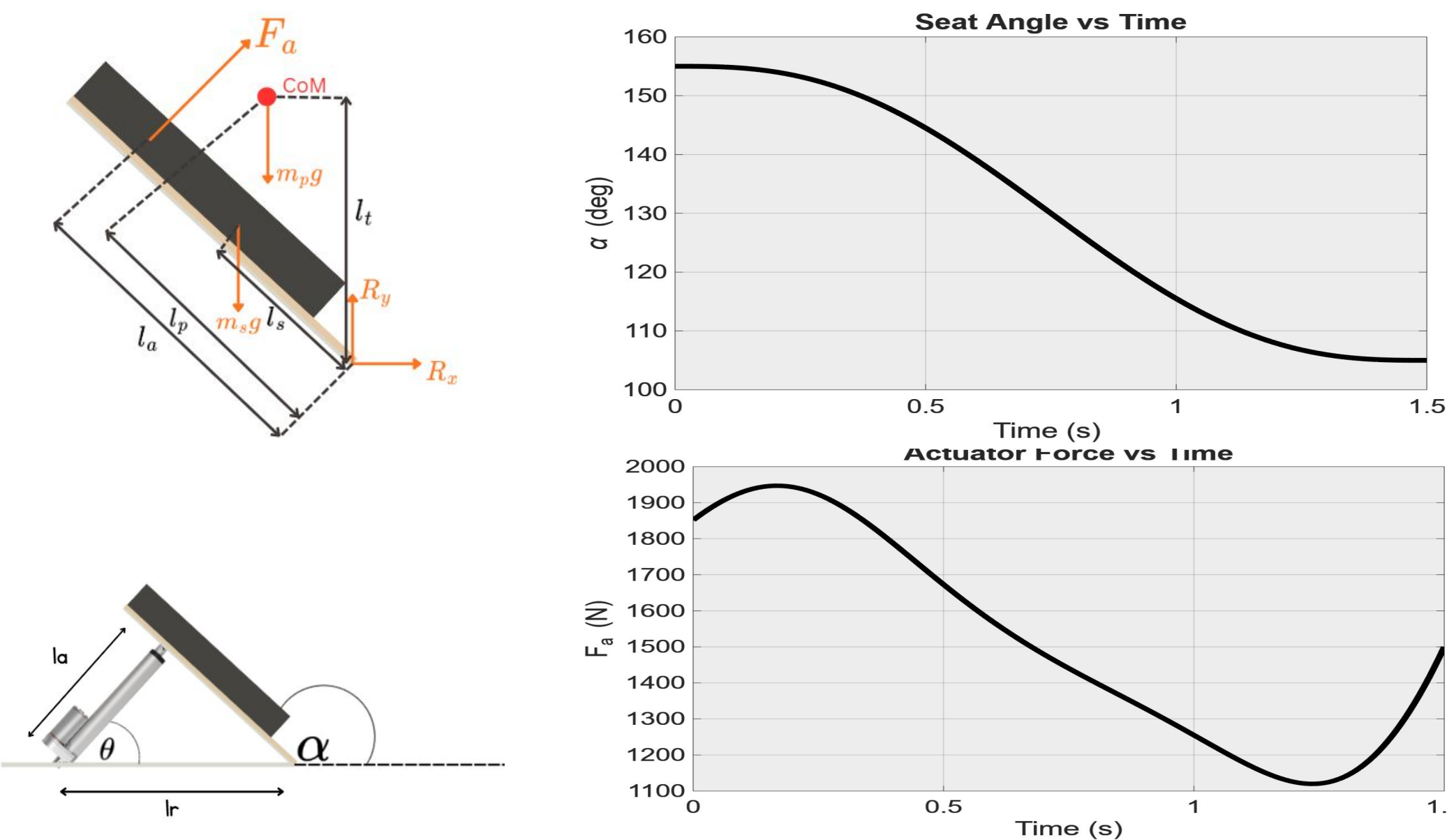
Tibault Dary-Alabaster, Zhenyu Hu, Darius Nguepi, Michael Rubin, Steph Akakabota

Low Fidelity Prototype

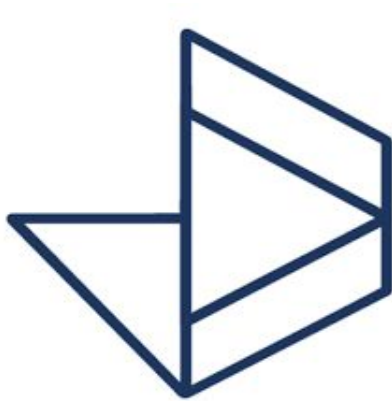
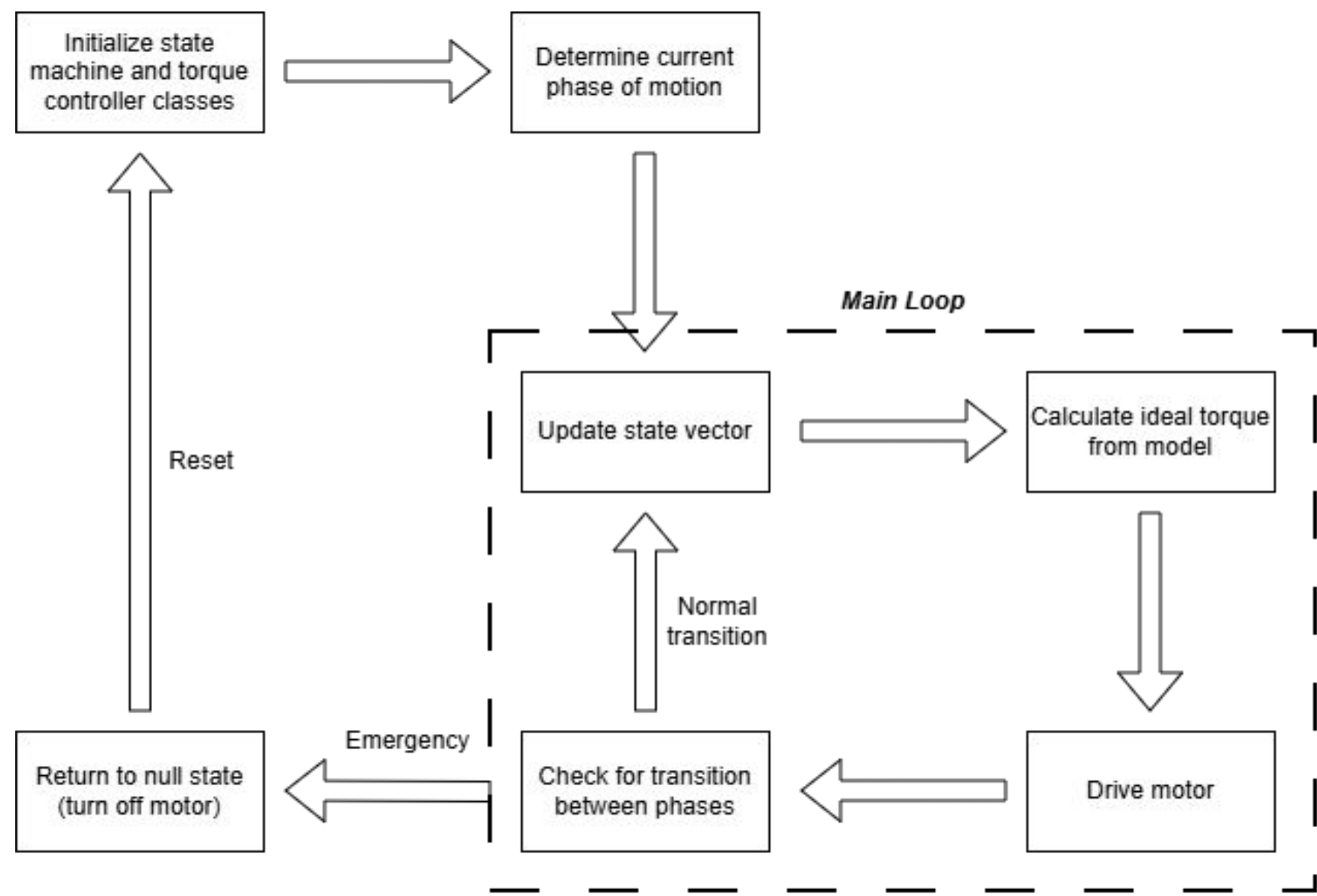


The low fidelity prototype validated the basic concept of SproutUp, confirming that a wearable sit-to-stand device was possible. Testing revealed key design improvements and next steps—decreasing strap length, increasing cushion size, and using more secure fastening methods.

Controls & Dynamics



Single-DOF kinematic model of idealized knee angle trajectory from literature (above). Used angle trajectory and dynamic model for torque control.



Final Prototype

Lightweight (4kg), powerful, comfortable. Adaptive assistance to match user need. Autonomous actuation based on sensor readings. Easy disengagement and storage when not in use. Backpack style straps and wearability.

